Biochemical structure of cell

- **Complete:**
    1. **Phospholipids** are the main type of lipid in cell membrane. (Specifically, Lecithins are described as the most important phospholipids of the cell membrane, and cholesterol is a major constituent).
    2. If the body weight of adult male is 75 kgm. The water content will be about **45 liters or kilograms**. Intracellular fluids will be about **30 L** (2/3 of the total water content), while extracellular fluid will be about **15 liters or kilograms** (1/3 of the total content).
- **Define:**

- **Apoptosis**: **Genetically regulated cell death**.
- **Malignancy**: cancar
- **Neoplasia** is **abnormal accumulation of cells**.
- **Prokaryotic Cell**: **Small, single cell lacking membrane-bound nucleus**.

- **Compare:**
    - Compare between prokaryote and Eukaryote?

Prokaryotes	**Eukaryotes**
Nuclear Membrane: **Has ill defined nuclear membranes**	Nuclear Membrane: **Well defined nuclear membranes**
Subcellular organelles: **No or less subcellular organelles**	Subcellular organelles: **Rich in subcellular organelles**

Examples: **Bacteria and blue-green algae**	Examples: **Yeast, fungi, protozoans, plants and animals**

PH of Blood, Acidosis, Alkalosis, Buffer systems and Extracellular fluids

- **Complete:**
    1. PH means: **-log[H+**
    2. The normal pH of the blood is **7.35 – 7.45 (Arterial)**.
    3. Acidosis of blood means pH is **decreases below 7.35**.
    4. Alkalosis of blood means pH is **exceeds 7.45**.
    5. PH of blood could be measured by **ABG apparatus**.
    6. PH of laboratory prepared solutions is measured by **pH Meter**.
- **True (√) or False (X) and mention the right answer (Justify):**
    - a. Acidosis means: high pH is above 7.5 (**X**). Justification: **Acidosis occurs when pH of the blood decreases below 7.35**. (pH above 7.45 is generally considered alkalosis).
    - b. Acidosis means: PH is below 7. 5 (**X**). Justification: **Acidosis occurs when pH of the blood decreases below 7.35**. (pH below 7.5 includes the normal range of 7.35-7.45).
    - c. Normal PH of arterial blood is between 7.35 – 7.45 (**√**). Justification: **Normal values of Arterial Blood Gases (ABG): PH 7.35 – 7.45**.
    - d. Chronic obstructive lung disease (as Asthma) is complicated with acidosis. (**√**). Justification: **Asthma is listed as a cause of respiratory acidosis**.
    - e. Chronic cigarette smoking usually leads to respiratory acidosis. (**√**). Justification: **Emphysema and chronic bronchitis**, conditions commonly associated with chronic smoking, **are listed as causes of respiratory acidosis**.

Amino Acids, polypeptides and proteins classifications

- **Definitions:**

- **Denaturation of proteins**: **Destruction of protein structures** (secondary, tertiary, quaternary).
- **High biological value proteins**: Proteins with **all essential amino acids** and **easily digested**.
- **Essential amino acids**: Amino acids **not synthesized in the body**; must be in diet.
- **Fibrous proteins**: Proteins with **axial ratio > 10**.
- **Enumerations:**
  1. **Functions of proteins**:
     - Nutritional.
     - Catalytic (Enzymes).
     - Hormonal (Hormones/receptors).
     - Immunological (Antibodies).
     - Structural (e.g. collagen).
     - Maintain blood osmotic pressure (Plasma proteins).
  2. **Effects of denaturation on proteins**:
     - Loss of catalytic activity of enzymes.
     - Loss of biological functions of hormones.
     - Decreased solubility.
     - Increased viscosity.
     - Increased digestibility.
  3. **Five of essential amino acids**
     - Isoleucine
     - Lysine
     - Histidine
     - Tryptophan
     - Methionine

4. **Metabolic classification subtype of amino acids**:
    - Glucogenic amino acids.
    - Ketogenic amino acids.
    - Mixed amino acids.
5. **Five types of neutral amino acids**
    - Aliphatic amino acids
    - Hydroxy amino acids
    - Aromatic amino acids
    - Sulfur-containing amino acids

- Heterocyclic amino acids**Four orders of structural organization of proteins**:

- primary,
- secondary,
- tertiary,
- Quaternary

II- Match the following statements in first column with the suitable answer in the second column

A	The answer
1- It consists of amylose and amylopectin	c. Starch
2- It is composed of 2 alpha glucose units united by alpha 1-4 glycosidic linkage.	f. Maltose
3- Is important for synthesis of lactose of milk.	a. Galactose
4- It is hydrolyzed by sucrase enzyme to glucose and fructose	b. Sucrose
5- Cannot be digested by humans	d. Cellulose
6- Is widely distributed among various animals and tissues including synovial fluid, the vitreous humor of the eye and loose connective tissues.	e. Hyaluronic acid

III- Put the suitable answer infront of every definition

1. They are repeating disaccharides units which are characterized by their content of amino sugars and uronic acids (**Glycosaminoglycans / Mucopolysaccharides**).
2. They are compounds having different spatial configuration, but have the same chemical formula (**Isomers**).
3. They are formed of more than 10 monosaccharides linked by glycosidic linkage and give monosaccharides on hydrolysis (**Polysaccharides**).
4. …..is a storage form of carbohydrates in liver and muscle (**Glycogen**).
5. Include compounds that are needed in amounts of a gram or more per day (**Macronutrients**).

IV- Complete the following statements with the suitable word.

1. **Chondroitin sulphate** is heteropolysaccharide that have a role in compressibility of cartilage and weight bearing protecting of joints. (Note: The source identifies Chondroitin sulphate as a heteropolysaccharide and Glycosaminoglycans as associated with structural elements, but does not explicitly detail this specific role).
2. All monosaccharides have to be converted into **glucose** to be Utilized by most of human tissues.
3. **Glucose** is the most important dietary of carbohydrate and the major metabolic fuel to the body.
4. Micronutrients generally function as **coenzymes** in metabolic reactions. (Note: The source explicitly links Vitamin B2, a micronutrient, and pentoses from carbohydrates to coenzymes, but does not generalize this function to all micronutrients).
5. Proteins yield **4 Kcal** cal/ gram. (Note: The source states the caloric yield for protein but does not explicitly state it for carbohydrates).

V- Define the following :

a. **Isomers in monosaccharides**: Compounds with the **same formula** but **different spatial arrangements**.

b. Examples about isomers in hexosugars. (has 6 carbons): **Glucose, Galactose, Fructose, Mannose**.

c. **Epimers**: Isomers differing in the **-OH/H configuration** on **one specific carbon**.

d. **Heparin**: A heteropolysaccharide, often **N-sulfated**, with **anticoagulant properties**. (Note: Source uses "Heparin" not "Heparin sulphate").

e. **Hyaluronic acid**: A heteropolysaccharide found in **synovial fluid, eye, connective tissues**.

VI- Enumerate :

1. **Four functions for glycoproteins:**

- Cell recognition
- Hormone receptors
- Structural support in membranes
- Immune response

II- Put the suitable answer in front of every definition:

1. Consist of phosphatidic acid esterified with ethanolamine **(Cephalins)**).
2. **(lung surfactant)**) is a very effective surface active agent, preventing adherence of the inner surfaces of the lungs.
3. This type of rancidity is caused by the action of certain fungi, producing ketones (**Ketonic rancidity**).
4. One of eicosanoids, derived from arachidonic acid by the action of Cyclooxygenase enzyme (**Prostaglandins (PGs)**).

5. Fatty acids that cannot be synthesized in the body at a sufficient rate and must be taken in diet (**essential fatty acids**).

III- Complete the following statements with the suitable word:

1. Estrogens and anrogens are derived from **Steroids**.
2. **Lecithins (Phosphatidyl cholines)** are the most important phospholipids of the cell membrane.
3. **Hydrolytic rancidity** caused by bacterial enzymes which cause hydrolysis of fat liberating free fatty acids.
4. Cod liver oil is rich in **ω-3 polyunsaturated** fatty acids.
5. **Glycerol** is a trihydric alcohol which can be esterified with one two or three fatty acids giving mono-, di-, or triglycerides respectively.

IV- Define the following:

a. **Essential fatty acids:** Fatty acids that cannot be synthesized in the body at a sufficient rate and must be taken in diet.

b. **Prostaglandins:** One of eicosanoids, derived from arachidonic acid by the action of Cyclooxygenase enzyme (COX). They act as local hormones.

c. **Lung surfactant:** Dipalmitoyl-lecithin, a very effective surface active agent preventing adherence of the inner surfaces of the lungs.

d. **Rancidity:** The development of bad odor and taste in the fat.

e. **Lecithin:** Phosphatidyl cholines, consisting of phosphatidic acid esterified with choline. They are the most important phospholipids of the cell membrane.

V. Enumerate:

1. Three importance of lipids:

- An efficient source of energy.
- Serve as thermal insulators.
- Act as electrical insulators.

2. Two glycerol-containing phospholipids:
    - Phosphatidyl cholines (Lecithins).
    - Phosphatidyl ethanolamine (Cephalins).
    - Phosphatidic acid.
3. Three types of rancidity:
    - Hydrolytic rancidity.
    - Ketonic rancidity.
    - Oxidative rancidity.
4. Three essential fatty acids:
    - Linoleic acid.
    - Linolenic acid.
    - Arachidonic acid.
5. Three unsaturated fatty acids:
    - Palmitoleic acid.
    - Oleic acid.
    - Linoleic acid.
    - Linolenic acid.
    - Arachidonic acid
6. Three phospholipids:
    - Phosphatidic acid.
    - Phosphatidyl cholines (Lecithins).
    - Phosphatidyl ethanolamine (Cephalins).
7. Three classes of lipids:
    - Simple lipids.
    - Complex or compound lipids.
    - Derived lipids (as steroids).

Vitamins role in our bodies Nutrition

8. **II- Match the following statements in column (A) with the suitable answer in the column (B)**

A	The answer
1- It is considered as natural antioxidants	d. Vitamin E
2- Its deficiency causes rickets	a. Vitamin D
3- Its coenzyme is FAD & FMN	e. Vitamin B2
4- It needs intrinsic factor for its absorption	b. Vitamin B12
5- It is given prophylactically in pregnancy to prevent spina bifida	f. Folic acid
6- Its deficiency causes scurvy	c. Vitamin C

III- Put true or false and correct the false statement

1. **True** – Vitamins are organic nutrients required in small quantities for normal growth and health.
2. **False** – The active form of vitamin D is **1,25-dihydroxycholecalciferol**.
3. **False** – Eye manifestations of vitamin B12 deficiency are not night blindness; **night blindness is due to vitamin A deficiency**.
4. **True** – Vitamin K2 is produced by intestinal bacteria.
5. **False** – Deficiency of vitamin E does **not** produce scurvy; **scurvy is caused by vitamin C deficiency**.
6. **False** – Hypernatremia means an **increase in sodium**, not potassium.

7. **False** – Hypokalemia means a **decrease in potassium**, not natrium (sodium).

IV- Complete:

1. Thiamine pyrophosphate is the coenzyme of vitamin **B1 (Thiamine)**.
2. Pellagra is characterized by **diarrhea, dermatitis, dementia**.
3. The coenzymes derived from niacin are **NAD** and **NADP**.
4. Thiamin deficiency causes **beriberi**.
5. Riboflavin is the scientific name of **vitamin B2**.

V- Put the suitable answer in front of the following statements:

1. The vitamin that has role in biosynthesis of clotting factors including (2,7,9,10) – **Vitamin K**.
2. The vitamin used in formation of CoA-SH and Acyl carrier protein – **Vitamin B5 (Pantothenic acid)**.
3. The vitamin that provides coenzyme PLP – **Vitamin B6 (Pyridoxine)**.
4. A glycoprotein secreted by the parietal cells of gastric mucosa and has role in B12 absorption – **Intrinsic factor**.
5. Type of anemia due to folate deficiency – **Megaloblastic anemia**.

VII- Short answered questions:

1. **Compare between fat & water-soluble vitamins**:
    - Fat-soluble vitamins (A, D, E, K) are stored in fat tissues and the liver, absorbed with dietary fat, and can accumulate to toxic levels.

- Water-soluble vitamins (B-complex, C) are not stored, need regular intake, and excess is excreted in urine.

2. **Enumerate three manifestations of vitamin A deficiency**:
   - Night blindness
   - Xerophthalmia
   - Keratomalacia
3. **List three sources of vitamin K**:
   - Green leafy vegetables (e.g., spinach, kale)
   - Liver
   - Intestinal bacteria

VIII-Define (Simplified & Short):

- **Scurvy**:
  Vitamin C deficiency causing gum bleeding and slow wound healing.
- **Pellagra**:
  Niacin deficiency causing diarrhea, skin issues, and memory loss.
- **Pernicious anemia**:
  Vitamin B12 deficiency due to lack of intrinsic factor.
- **Spina Bifida**:
  A birth defect where the spine doesn’t close properly.

Minerals/Nutrition (Implicit Section)

II- Complete statements:

1. Main intracellular cation is **potassium (K^+)**.
2. Main extracellular cation is **sodium (Na^+)**.

3. Deficiency of potassium is called **hypokalemia**.
4. Deficiency of sodium is called **hyponatremia**.
5. Body Mass Index (BMI) means **weight in kilograms divided by height in meters squared**.
6. Waist to hip ratio normal in male equals **≤ 0.9**.
7. Normal random blood sugar should be less than **140 mg/dL**.

Feature	Plasma	Serum
Anticoagulant	Required to prevent clotting. Examples: EDTA, heparin, citrate.	Not required, as blood is allowed to clot.
How to Obtain	Collect blood in a tube with anticoagulant. Centrifuge to separate.	Collect blood in a tube without anticoagulant. Allow to clot. Centrifuge to separate.
Uses	- Coagulation studies	- Immunological tests (antibody detection)
-	- Plasma Proteins transfusions - Biochemical analyses (electrolyte, enzyme levels)	
- Measuring certain medications	- Hormone assays	
- Research	- Research	